

# EXACT ASYMPTOTIC EFFICIENCY UNDER ZCDP: DIAMETER-CONSTRAINED INFORMATION GEOMETRY

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Protecting individual privacy has become a central and urgent concern in modern data analysis, given the vast quantities of data now generated and processed. In this paper, we develop a systematic theory of exact asymptotic efficiency for regular parametric estimation under central zero-concentrated differential privacy (zCDP). In the privacy regime, the governing object is a diameter-constrained information region: the set of information matrices generated by statistics with diameter at most one. For weighted quadratic loss, we show that the exact local minimax risk is an inverse information variational functional over this region.

More generally, a mixed information region yields a unified efficiency constant across different regimes, covering the privacy regime and classical Fisher efficiency. A matching estimator releases the empirical mean of a nearly optimal bounded statistic with Gaussian noise and locally inverts its population moment map. Our theory differs from classical efficiency theory in its set-valued information geometry and loss-dependent efficient estimator.

As examples, we provide closed-form constants and optimal procedures for various concrete models, including one-dimensional regular families, Gaussian means, categorical probability vectors, and regression models among others. The theory also yields exact asymptotic efficiency under Gaussian differential privacy through a precise parameter rescaling.

**1. Introduction.** Differential privacy (DP) provides a principled framework for statistical analysis while limiting the disclosure of individual records (Dwork et al., 2006; Dwork and Roth, 2014). As data increasingly originate from sensitive sources, ranging from medical records and financial transactions to personal devices, the need for rigorous privacy protection has become both a scientific imperative and a legal requirement. Preserving privacy while enabling meaningful statistical inference is therefore one of the central challenges of modern data analysis.

Since the introduction of DP, numerous private methods have been developed for a broad range of estimation problems. In parallel, a large body of work has been devoted to establishing minimax optimal rates under privacy for specific problems, including mean estimation, regression, and distribution estimation (Duchi, Jordan and Wainwright, 2018; Acharya, Sun and Zhang, 2021; Cai, Wang and Zhang, 2021). These results characterize the order of the privacy cost, but these rate bounds do not determine the exact leading constant.

A central statistical question remains unresolved: how to characterize *efficiency* under DP, namely the *minimax optimal constant*. The key challenge is that privacy imposes a global constraint on the estimation procedure, not the statistical model itself, so the classical efficiency

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theory based on Fisher information does not simply apply. This leads to our central questions: what information geometry under privacy governs efficiency, what exact local minimax constant does it imply, and can that constant be attained by an explicit private estimator?

We answer this question by developing a general efficiency theory for private parametric estimation. Consider  $n$  independent observations from a regular parametric model indexed by  $\theta \in \Theta \subset \mathbb{R}^p$ . In this paper, we focus on central record-level zero-concentrated differential privacy (zCDP) (Dwork and Rothblum, 2016; Bun and Steinke, 2016) with privacy budget  $\rho_n$ , while our results also extend directly to Gaussian DP (Dong, Roth and Su, 2022). Beginning with the high privacy regime in which the privacy error dominates, we identify the exact asymptotic minimax constant over the entire class of private mechanisms and construct optimal estimators that attain it.

We find that the information geometry governing private estimation is fundamentally different from ordinary Fisher information. For example, consider scalar Gaussian mean estimation,  $N(\theta, \sigma^2)$ , whose ordinary Fisher information is  $\mathbb{E}_\theta s_\theta(X)^2 = \sigma^{-2}$ , where  $s_\theta(X) = (X - \theta)/\sigma^2$  is the score function. Under zCDP, the optimal private mechanism is not a Gaussian release of the sample mean, and the corresponding local minimax risk is not determined by the Fisher information. Instead, the optimal private mechanism is a noisy threshold count: we first count  $C_n = \sum_{i=1}^n \mathbf{1}\{X_i \geq \theta_0\}$  and privatize it as  $\tilde{C}_n = C_n + Z_n$  for  $Z_n \sim N(0, 1/(2\rho_n))$ ; then project  $\tilde{C}_n/n$  onto a fixed interval inside the image neighborhood of the mean map  $g_{\theta_0}(\theta) = P_\theta(X \geq \theta_0)$  and apply its local inverse. It can be shown that the resulting private risk is governed by the squared  $L^1$  score quantity  $(\mathbb{E}_{\theta_0} |s_{\theta_0}(X)|/2)^2 = (2\pi\sigma^2)^{-1}$ , and this risk is in fact the optimal local zCDP constant for this model; see the scalar specialization of Proposition 5.3. This example already shows that privacy changes the underlying local information geometry, not merely the effective sample size.

The preceding construction generalizes to a simple but fundamental class of private estimators, which we call *private moment-map inverse estimators*. Let  $\Phi : \mathcal{X} \rightarrow \mathbb{R}^p$  be a statistic with diameter at most one, that is,  $\text{diam}(\Phi) := \sup_{x, x' \in \mathcal{X}} \|\Phi(x) - \Phi(x')\|_2 \leq 1$ . Its empirical mean has replacement sensitivity at most  $1/n$ , so the Gaussian mechanism (Dwork and Rothblum, 2016; Bun and Steinke, 2016) gives a  $\rho_n$ -zCDP release

$$Y_{\Phi, n} = \frac{1}{n} \sum_{i=1}^n \Phi(X_i) + G_n, \quad G_n \sim N(0, (2n^2\rho_n)^{-1}I_p),$$

If the population moment map  $g_\Phi(\theta) = \mathbb{E}_\theta \Phi(X)$  is continuously differentiable with nonsingular derivative at  $\theta_0$ , the inverse function theorem gives a local inverse  $\Psi_\Phi$ . Fix a closed Euclidean ball  $\mathcal{B}_\Phi$  centered at  $g_\Phi(\theta_0)$  and compactly contained in the domain of  $\Psi_\Phi$ , and let  $\Pi_{\mathcal{B}_\Phi}$  denote Euclidean projection onto it. We estimate  $\theta$  by the projected local inversion

$$\hat{\theta}_{\Phi, n} = \Psi_\Phi(\Pi_{\mathcal{B}_\Phi} Y_{\Phi, n}).$$

We call  $\hat{\theta}_{\Phi, n}$  the private moment-map inverse estimator based on  $\Phi$ . Consequently, the private noise  $G_n$  contributes to the error through the inverse of the relevant Jacobian

$$A_{\Phi, \theta_0} := Dg_\Phi(\theta_0) = \mathbb{E}_{\theta_0} [\Phi(X) s_{\theta_0}(X)^\top],$$

or more precisely, through the induced positive semidefinite information matrix  $A_{\Phi, \theta_0}^* A_{\Phi, \theta_0}$ , where  $A_{\Phi, \theta_0}^*$  is the adjoint of  $A_{\Phi, \theta_0}$ .

Motivated by this construction, we introduce the *diameter-constrained information region*

$$\mathcal{J}_{\theta_0} = \text{cl}\{A_{\Phi, \theta_0}^* A_{\Phi, \theta_0} : \text{diam}(\Phi) \leq 1\},$$

where  $\Phi$  ranges over diameter-one statistics. We refer to Definition 2.2 for a precise definition. Rather than selecting a single maximal information matrix,  $\mathcal{J}_{\theta_0}$  reflects the local

TABLE 1  
Asymptotic regimes and corresponding results under zCDP.

|          | Strong privacy      | Privacy dominated                                       | Transition                                                                            | Sampling dominated                     |
|----------|---------------------|---------------------------------------------------------|---------------------------------------------------------------------------------------|----------------------------------------|
| Relation | $n^2 \rho_n = O(1)$ | $n^2 \rho_n \rightarrow \infty, n \rho_n \rightarrow 0$ | $n \rho_n \rightarrow \lambda \in (0, \infty),$<br>$\beta := \lambda / (1 + \lambda)$ | $n \rho_n \rightarrow \infty$          |
| Error    | $\geq c > 0$        | $(n^2 \rho_n)^{-1} \mathfrak{C}_{\theta_0}(W)$          | $n^{-1} \mathfrak{C}_{\theta_0}^{(\beta)}(W) / \beta$                                 | $n^{-1} \text{Tr}[W I(\theta_0)^{-1}]$ |

information matrices generated by all such statistics. Our minimax lower bound shows that the same region constrains the information of every zCDP output experiment, and hence the first-order risk of every zCDP estimator. Its matrix-valued nature is essential in several dimensions, where the optimal statistic depends on the loss and on how information is allocated across parameter directions.

For a quadratic loss matrix  $W \succeq 0$ , define

$$\mathfrak{C}_{\theta_0}(W) = \frac{1}{2} \inf_{J \in \mathcal{J}_{\theta_0}} \mathcal{I}_W(J),$$

where  $\mathcal{I}_W(J) = \text{Tr}(W J^{-1})$  for nonsingular  $J$ ; the precise definition for all  $J \succeq 0$  is given in [Definition 2.5](#). Under the privacy regime and the regularity conditions, our main theorem establishes the exact local minimax risk:

$$(1) \quad \lim_{n \rightarrow \infty} n^2 \rho_n \inf_{\hat{\theta}: \rho_n\text{-zCDP}} \sup_{\theta \in \Theta_n} \mathbb{E}_\theta \|\hat{\theta} - \theta\|_W^2 = \mathfrak{C}_{\theta_0}(W),$$

where  $\Theta_n$  is a shrinking neighborhood of  $\theta_0$  and the infimum is over all zCDP estimators. The optimal constant is attained by private moment-map inverse estimators based on nearly optimal  $\Phi$  from the information region. In addition, we show the asymptotic normality of the attaining estimator. For  $W \succ 0$ , let  $J_W$  be the unique optimizing information matrix, and an attaining sequence can be chosen so that the law of  $\tau_n(\hat{\theta}_n - \theta)$  converges uniformly over the local neighborhood to  $N(0, J_W^{-1}/2)$ .

A distinctive feature of the private efficiency bound is that the optimizing information matrix, and hence the choice of an efficient diameter-one statistic, can depend on the loss matrix  $W$ . If  $J_1 \succeq J_2$ , then  $\mathcal{I}_W(J_1) \leq \mathcal{I}_W(J_2)$  for every  $W \succeq 0$ . However,  $\mathcal{J}_{\theta_0}$  need not possess a greatest element in the Loewner order, so there need not be a single feasible information matrix, or a single statistic  $\Phi$ , that is optimal for every loss. [Proposition 2.15](#) shows that no single zCDP estimator sequence can attain both loss-specific constants in this case. This contrasts with classical regular efficiency, where the same Fisher information matrix governs every quadratic loss and a single asymptotically efficient estimator—such as the maximum likelihood estimator—attains the efficiency bound for every fixed  $W$ .

Moreover, our theory also characterizes efficiency across the different asymptotic regimes, rather than only the privacy-dominated regime. [Table 1](#) summarizes these regimes and their risk behavior. When  $n^2 \rho_n \rightarrow \infty$ , a unified scale and mixed information region yield exact efficiency constants, retaining both the sampling and privacy covariance of the same statistic; see [Theorems 2.8](#) and [2.12](#). In particular, when the privacy budget is large, the efficiency constant is just the ordinary Fisher constant. When the privacy cost is comparable to the statistical cost, we characterize efficiency through a joint sampling–privacy variational constant in [Definition 2.7](#). On the other hand, in the strong privacy regime where  $n^2 \rho_n = O(1)$ , consistency is impossible; see [Proposition 2.11](#).

Our efficiency theory applies to arbitrary fixed-dimensional regular parametric models under zCDP and weighted quadratic losses. As applications, we obtain exact constants in

closed form and construct matching private estimators for several important classes of models. These include one-dimensional regular families, Gaussian and Poisson means, categorical probability vectors, bounded mean estimation, and generalized linear regression under Gaussian design. The Supplementary Material (Cai and Li, 2026a) gives a complete catalogue of further exact constants, including classical location and scale families, spherically symmetric location models, and additional regression settings. These examples illustrate the broad applicability of our theory and the diversity of information geometries that arise under privacy.

Finally, our efficiency results extend exactly to  $\mu_n$ -Gaussian differential privacy (GDP) (Dong, Roth and Su, 2022). Upon setting  $\rho_n = \mu_n^2/2$ , the same effective scale and optimal constants apply, since every  $\mu_n$ -GDP mechanism is  $\rho_n$ -zCDP, while the Gaussian releases attaining the zCDP bounds are themselves  $\mu_n$ -GDP.

**1.1. Related work.** The local minimax formulation follows the classical strategy of deriving an information bound and constructing a procedure that attains it. Our lower bound uses a matrix van Trees inequality, following the multivariate Bayesian Cramér–Rao formulation of Gill and Levit (1995). The new issue created by privacy is not the van Trees inequality itself, but the characterization of the Fisher information matrices attainable by a private output experiment.

**1.1.1. Efficiency and exact optimality under local DP.** Local DP (LDP) requires each observation to be privatized before aggregation, so its mechanisms have a simpler structure than central DP mechanisms, which can process the entire sample jointly. For regular parametric models under a fixed LDP level, Steinberger (2024) establishes local asymptotic mixed normality for arbitrary sequentially interactive mechanisms and derives convolution and local asymptotic minimax results. For scalar parameters, the paper identifies the minimal asymptotic variance as the reciprocal of the maximal Fisher information over marginal LDP channels and shows that a two-step private MLE attains it. For multidimensional parameters, the convolution and local asymptotic minimax bounds are stated in terms of the Fisher information of the privatized experiment, without identifying an optimal information matrix or optimal constant.

Within this scalar LDP framework, Kalinin and Steinberger (2025) solve the channel optimization in closed form for the unit-variance Gaussian location model when  $\epsilon \leq 1.04$ : randomized response applied to the sign of  $X - \theta$  yields Fisher information  $\frac{2}{\pi} \left( \frac{e^\epsilon - 1}{e^\epsilon + 1} \right)^2$ , and a two-stage estimator attains the reciprocal variance. However, their results are restricted to the Fisher information for the Gaussian model. Recent work further studies extremal and staircase representations for Fisher information and privacy–utility optimization under LDP (Amorino and Gloter, 2025, 2026).

For a distribution on a fixed  $k$ -point alphabet under noninteractive  $\epsilon$ -LDP, Ye and Barg (2019) show that, for every  $1 \leq p \leq 2$ , the worst-case  $\ell_p^p$  risk of their scheme and estimator divided by the minimax risk converges to one as the sample size grows. This exact result is confined to global worst-case risk for a fixed finite alphabet model with fixed  $\epsilon$  and noninteractive observation-wise privatization.

**1.1.2. Optimal rates and information contraction under privacy.** A substantial literature studies how privacy changes minimax rates and develops lower-bound methods for establishing those rates. Under local privacy, Duchi, Jordan and Wainwright (2018) derive private divergence bounds and rate-optimal procedures for several estimation problems, while Rohde and Steinberger (2020) characterize private rates through moduli of continuity and construct rate-optimal channels. At the instance-specific level, Duchi and Ruan (2024) show that LDP

local minimax risk is governed by a total-variation modulus and define, for one-dimensional regular models, the  $L^1$ -information  $J_\theta = \mathbb{E}_\theta |s_\theta(X)|$ , equal to  $2a_\theta$  in our scalar notation. Their lower bound covers sequentially interactive locally Rényi-private channels, and matching constructions recover this local risk scale in several settings, but the comparison is only up to universal numerical constants rather than an exact leading constant. Under central DP, [Smith \(2011\)](#) constructs private versions of a broad class of estimators that retain their classical asymptotic distributions, and [Awan and Cai \(2025\)](#) preserve efficient summary statistics in differentially private synthetic data. For minimax risk under central DP, [Cai, Wang and Zhang \(2021\)](#) establish matching rates for mean estimation and linear regression in low- and high-dimensional settings, and [Acharya, Sun and Zhang \(2021\)](#) develop private analogues of Le Cam, Fano, and Assouad methods. Building on tracing attacks, [Cai, Wang and Zhang \(2023\)](#) use model scores to construct a general attack statistic and obtain privacy-constrained minimax lower bounds and optimal rates up to logarithmic factors for several parametric and nonparametric models. These results characterize the order of the privacy cost, or provide general tools for doing so, without by themselves identifying a loss-specific leading local constant.

Fisher information contraction offers a more local route to private lower bounds. [Barnes, Han and Özgür \(2019\)](#) study Fisher information under blackboard communication protocols, and [Barnes, Chen and Özgür \(2020\)](#) derive Fisher information bounds under local differential privacy. Under central zCDP, [Cai and Li \(2026b\)](#) prove a trace contraction for the Fisher information of any  $\rho$ -zCDP statistic and combine it with a multivariate van Trees inequality. However, their contraction does not yield a sharp constant and fails to capture how the optimal constant depends on the loss.

*1.1.3. Optimal constants and fixed-query mechanisms.* A related line of work fixes a query and optimizes its release mechanism. For count queries, [Ghosh, Roughgarden and Sundararajan \(2009\)](#) establish the universal optimality of the geometric mechanism for their class of users and losses. For real-valued queries with bounded sensitivity under pure differential privacy, [Geng and Viswanath \(2014\)](#) characterize optimal staircase noise. [Geng et al. \(2019\)](#) study optimal noise addition under additive differential privacy, and [Kulesza, Suresh and Wang \(2025\)](#) develop general staircase mechanisms. Under  $f$ -DP, [Awan and Ramasethu \(2024\)](#) optimize additive noise through anti-concentration and stochastic dominance. These formulations optimize a release channel after the query and its sensitivity geometry have been specified. A population estimation problem additionally requires choosing which statistic should encode the model score and proving a lower bound over all admissible mechanisms.

Recent work has used staircase mechanism arguments to claim exact constants for one-dimensional mean estimation and for variance and covariance estimation ([Kulesza, Suresh and Wang, 2024](#); [Takakura, Liew and Hasegawa, 2026](#)) under high privacy. However, fixed-query staircase characterizations do not by themselves yield lower bounds over arbitrary central DP mechanisms.

*1.2. Organization.* [Section 2](#) formalizes the local zCDP experiment, introduces the diameter-constrained and mixed information regions, and states the privacy-dominated and unified local and compact efficiency theorems. [Section 3](#) constructs estimators that attain the bound, and [Section 4](#) proves the matching lower bound over all zCDP mechanisms by combining uniform information contraction with a matrix local minimax argument. [Section 5](#) develops closed-form constants for several model classes. [Section 6](#) treats Gaussian differential privacy and differentiable targets, and [Section 7](#) summarizes the implications, limitations, and remaining questions. The supplement contains the complete technical proofs and further example calculations.

Write  $\mathbb{S}^p$  for the space of real symmetric  $p \times p$  matrices and  $\mathbb{S}_+^p$  for its positive semidefinite cone. Use  $\preceq$  and  $\prec$  to denote the Loewner order on  $\mathbb{S}^p$ . Denote the  $k \times k$  identity matrix by  $I_k$ . Write  $A^\top$  for the transpose of a matrix and  $A^*$  for the adjoint of a linear operator. For any matrix  $A$ ,  $A^\dagger$  denotes its Moore–Penrose inverse. We use  $\text{Tr}$  for the trace and  $\|\cdot\|_2$  for the Euclidean norm, with other norms indicated by their subscripts. For  $W \succeq 0$ , write  $\|v\|_W = (v^\top W v)^{1/2}$  for the induced seminorm.

## 2. Efficiency under zCDP.

**2.1. Setup.** Let  $\{P_\theta : \theta \in \Theta\}$  be a parametric model on a standard Borel sample space  $\mathcal{X}$ , where  $\Theta \subset \mathbb{R}^p$  is open and  $p$  is fixed. Suppose that the model is dominated by a common  $\sigma$ -finite measure  $\mu$ , with densities  $p_\theta$ . We have independent observations  $X_1, \dots, X_n$  from  $P_\theta$ .

Throughout this paper, we adopt the notion of  $\rho$ -zCDP from [Bun and Steinke \(2016\)](#), which is a convenient relaxation of  $(\epsilon, \delta)$ -DP. We use the Rényi divergence of order  $\alpha > 1$  to quantify the distinguishability of two distributions  $P$  and  $Q$ :  $D_\alpha(P\|Q) = \frac{1}{\alpha-1} \log \int \left(\frac{dP}{dQ}\right)^\alpha dQ$ . We say that two data sets  $S, S' \in \mathcal{X}^n$  are adjacent if they differ in one coordinate, i.e., if there exists an index  $i$  such that  $S_j = S'_j$  for all  $j \neq i$  and  $S_i \neq S'_i$ . The following definition is from [Bun and Steinke \(2016\)](#); [Dwork and Rothblum \(2016\)](#).

**DEFINITION 2.1** ( $\rho$ -zCDP). A randomized algorithm  $M : \mathcal{X}^n \rightarrow \mathcal{Z}$  is  $\rho$ -zCDP if, for every adjacent pair  $S, S' \in \mathcal{X}^n$ ,

$$D_\alpha(M(S)\|M(S')) \leq \alpha\rho, \quad \forall \alpha > 1.$$

The divergence formulation of zCDP is more convenient and elegant for our efficiency analysis than the probability formulation of  $(\epsilon, \delta)$ -DP, though the latter is more common in the literature. Moreover, it is well known that  $\rho$ -zCDP implies  $(\epsilon, \delta)$ -DP for any  $\delta \in (0, 1)$  with  $\epsilon = \rho + 2\sqrt{\rho \log(1/\delta)}$ , while  $(\epsilon, 0)$ -DP implies  $(\epsilon^2/2)$ -zCDP ([Bun and Steinke, 2016](#)). However, these translation formulas may not be tight, and the exact efficiency constants under  $(\epsilon, \delta)$ -DP may differ from those under zCDP. Gaussian differential privacy admits an exact transfer because every  $\mu$ -GDP mechanism is  $(\mu^2/2)$ -zCDP and the Gaussian mechanisms used here are themselves  $\mu$ -GDP; see [Subsection 6.1](#). We will discuss possible extensions of our results to  $(\epsilon, \delta)$ -DP in [Section 7](#).

We impose two basic regularity conditions on the statistical model. The first is classical regularity assumption ([van der Vaart, 1998](#)) ensuring the existence of score functions and Fisher information. The second ensures that the score functions are well-behaved in  $L^1$  so that they can be used to define our diameter-constrained information for private estimation.

**ASSUMPTION 1** (DQM regularity). The model is differentiable in quadratic mean (DQM) at every  $\theta \in \Theta$  with score  $s_\theta$  for one observation satisfying

$$\mathbb{E}_\theta s_\theta(X) = 0, \quad \mathbb{E}_\theta \|s_\theta(X)\|_2^2 < \infty.$$

**ASSUMPTION 2** ( $L^1$  smoothness). The density map  $\theta \mapsto p_\theta$  is continuously differentiable as an  $L^1(\mu)$ -valued map on  $\Theta$ , with derivative  $\dot{p}_\theta(x) = p_\theta(x)s_\theta(x)$ .

[Assumptions 1](#) and [2](#) are mild regularity conditions that hold for many parametric models. In particular, both assumptions hold for regular natural exponential families with common support and an open natural parameter space. Common examples include the Gaussian location family, Bernoulli family, and Cauchy location family among others.

2.2. *Diameter-constrained private information.* The classical theory of asymptotic efficiency is based on the Fisher information matrix

$$I(\theta) := \mathbb{E}_\theta \left[ s_\theta(X) s_\theta(X)^\top \right],$$

which defines the local information metric of the statistical model and quantifies the infinitesimal distinguishability of nearby distributions under full observation. Under privacy, however,  $I(\theta)$  alone does not characterize the information attainable by the output experiment, because privacy restricts the class of statistics that can be released. More concretely, if we use the Gaussian mechanism (see [Lemma 3.1](#)), then we can only transmit information through a statistic with bounded diameter. This motivates the following diameter-constrained information region.

Suppose [Assumption 1](#) holds, and fix  $\theta \in \Theta$ . For a real Hilbert space  $\mathcal{H}$  and a statistic  $\Phi : \mathcal{X} \rightarrow \mathcal{H}$  satisfying  $\mathbb{E}_\theta \|\Phi(X)\|_{\mathcal{H}}^2 < \infty$ , the diameter of  $\Phi$  is

$$(2) \quad \text{diam}(\Phi) := \sup_{x, x' \in \mathcal{X}} \|\Phi(x) - \Phi(x')\|_{\mathcal{H}}.$$

We introduce the score–statistic covariance operator

$$(3) \quad A_{\Phi, \theta} : \mathbb{R}^p \rightarrow \mathcal{H}, \quad A_{\Phi, \theta} h = \mathbb{E}_\theta [\Phi(X) \langle s_\theta(X), h \rangle] \quad h \in \mathbb{R}^p.$$

The adjoint of  $A_{\Phi, \theta}$  is  $A_{\Phi, \theta}^* : \mathcal{H} \rightarrow \mathbb{R}^p$ , given by  $A_{\Phi, \theta}^* v = \mathbb{E}_\theta [s_\theta(X) \langle \Phi(X), v \rangle_{\mathcal{H}}]$ ,  $v \in \mathcal{H}$ .

DEFINITION 2.2. The diameter-constrained information region at  $\theta$  is

$$(4) \quad \mathcal{J}_\theta := \text{cl} \left\{ A_{\Phi, \theta}^* A_{\Phi, \theta} : d < \infty, \Phi : \mathcal{X} \rightarrow \mathbb{R}^d, \text{diam}(\Phi) \leq 1 \right\} \subset \mathbb{S}_+^p,$$

where the closure is taken in  $\mathbb{S}^p$ . We identify the positive semidefinite cone  $\mathbb{S}_+^p$  with the set of self-adjoint positive semidefinite operators on  $\mathbb{R}^p$ .

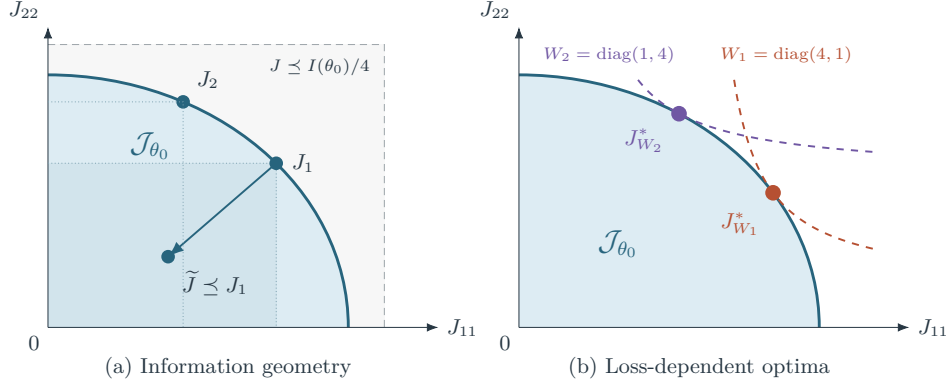
We emphasize that  $\mathcal{J}_\theta$  is a set of positive semidefinite matrices, not a single number or matrix. In comparison with the Fisher information, the geometry under privacy could be more complicated, and the information region may contain multiple extreme points, so a single matrix may not be sufficient. The set  $\mathcal{J}_\theta$  characterizes the information attainable by any statistic with diameter at most one, and it is the fundamental object that governs the local minimax risk under zCDP. We may reduce it by taking the trace to obtain a scalar quantity:

$$(5) \quad \Gamma_\theta := \sup_{J \in \mathcal{J}_\theta} \text{Tr } J = \sup_{d < \infty, \Phi : \mathcal{X} \rightarrow \mathbb{R}^d, \text{diam}(\Phi) \leq 1} \|A_{\Phi, \theta}\|_{\text{HS}}^2.$$

We next provide some basic geometric properties of  $\mathcal{J}_\theta$ . See also [Figure 1](#).

PROPOSITION 2.3. Suppose [Assumption 1](#) holds, and fix  $\theta \in \Theta$ . Then the following statements hold:

- (i) For every real Hilbert space  $\mathcal{H}$ , every statistic  $\Phi : \mathcal{X} \rightarrow \mathcal{H}$  with finite second moment, and every  $c \in \mathcal{H}$ ,  $\text{diam}(\Phi + c) = \text{diam}(\Phi)$ ,  $A_{\Phi+c, \theta} = A_{\Phi, \theta}$ .
- (ii)  $\mathcal{J}_\theta$  is a nonempty compact convex subset of  $\mathbb{S}_+^p$ .
- (iii) The region  $\mathcal{J}_\theta$  is downward closed in the Loewner order: if  $J \in \mathcal{J}_\theta$  and  $0 \preceq \tilde{J} \preceq J$ , then  $\tilde{J} \in \mathcal{J}_\theta$ .
- (iv) The region  $\mathcal{J}_\theta$  contains a positive definite matrix if and only if  $I(\theta) \succ 0$ .

FIG 1. Illustration of the information region  $\mathcal{J}_{\theta_0}$ .

The definition in (4) is characterized over all finite-dimensional statistics  $\Phi$ , but actually allowing arbitrary real Hilbert-valued statistics does not enlarge  $\mathcal{J}_\theta$ . Projection onto the range of  $A_{\Phi, \theta}$ , which has dimension at most  $p$ , preserves  $A_{\Phi, \theta}^* A_{\Phi, \theta}$  exactly while not increasing the diameter.

Moreover, the diameter-constrained information region is closely related to the local total variation distance between nearby distributions. For probability measures  $P, Q$  dominated by  $\mu$ , with densities  $p, q$ , respectively, write  $\text{TV}(P, Q) := \frac{1}{2} \int |p - q| d\mu$ .

**PROPOSITION 2.4.** Suppose [Assumptions 1](#) and [2](#) hold, and fix  $\theta \in \Theta$ . Then, for every  $h \in \mathbb{R}^p$ ,

$$(6) \quad \sup_{J \in \mathcal{J}_\theta} h^\top J h = \frac{1}{4} (\mathbb{E}_\theta |\langle s_\theta(X), h \rangle|)^2.$$

Moreover, as  $t \rightarrow 0$ ,

$$(7) \quad \text{TV}(P_{\theta+th}, P_\theta) = \frac{|t|}{2} \mathbb{E}_\theta |\langle s_\theta(X), h \rangle| + o(|t|) = |t| \sqrt{\sup_{J \in \mathcal{J}_\theta} h^\top J h} + o(|t|).$$

By (6) and the Cauchy–Schwarz inequality,

$$(8) \quad J \preceq \frac{1}{4} I(\theta) \quad \text{for every } J \in \mathcal{J}_\theta, \quad \Gamma_\theta \leq \frac{1}{4} \text{Tr } I(\theta).$$

Thus the Fisher information provides a universal quadratic upper bound, whereas the exact directional envelope is governed by the  $L^1$  norm of the directional score and need not itself be a quadratic form. In particular, different directions may require different score-sign statistics, which explains why the multivariate private information is naturally represented by a region rather than by a single maximal matrix.

When  $p = 1$ , we have the following simple characterization of the information region.

[Example 2.1](#) shows that the diameter-constrained information region is related to the  $L^1$  norm of the score, rather than by the Fisher information  $\mathbb{E}_\theta s_\theta(X)^2$ . This shows that our diameter-constrained information is substantially different from the Fisher information.

**EXAMPLE 2.1 (One-dimensional information region).** Under  $p = 1$ , we have

$$\mathcal{J}_\theta = [0, \Gamma_\theta], \quad \Gamma_\theta = \left( \frac{1}{2} \mathbb{E}_\theta |s_\theta(X)| \right)^2,$$

where the equality is attained by  $\phi_*(x) = \frac{1}{2} \text{sign}[s_\theta(x)]$ .

**2.3. Privacy-dominated local efficiency.** Our first result focuses on the privacy-dominated regime, where the cost of privacy dominates the statistical error. For a sequence of privacy budgets  $\rho_n$ , write  $\tau_n := n\sqrt{\rho_n}$ . For the privacy-dominated results, we work in the asymptotic regime where  $n \rightarrow \infty$  and

$$(9) \quad \rho_n \rightarrow 0, \quad \tau_n \rightarrow \infty, \quad n\rho_n \rightarrow 0.$$

We recall that the optimal rates (Cai, Wang and Zhang, 2021) of convergence under zCDP are often  $n^{-1} + \tau_n^{-2}$ . Therefore, the condition  $\rho_n \rightarrow 0$  describes an increasingly stringent privacy constraint;  $\tau_n \rightarrow \infty$  ensures consistency of the private estimator; while  $n\rho_n \rightarrow 0$  makes the statistical error negligible relative to the privacy noise.

We shall first consider a local estimation problem around a fixed center  $\theta_0 \in \Theta$ . Given any sequence  $R_n \rightarrow \infty$  with  $R_n/\tau_n \rightarrow 0$ , define

$$(10) \quad \Theta_n = \left\{ \theta_0 + \frac{u}{\tau_n} : \|u\|_2 \leq R_n \right\}.$$

Because  $\Theta$  is open,  $\Theta_n \subset \Theta$  for all sufficiently large  $n$ .

To state the theorem, we introduce a trace functional to accommodate singular matrices. For  $W \succeq 0$  and  $J \succeq 0$ , define  $\mathcal{I}_W(J) = \lim_{t \downarrow 0} \text{Tr}[W(J + tI_p)^{-1}]$ . If  $J$  is nonsingular, then  $\mathcal{I}_W(J) = \text{Tr}(WJ^{-1})$ . More generally, if  $\mathbb{R}^p = \text{ran}(J) \oplus \ker(J)$ , then  $\mathcal{I}_W(J) < \infty$  precisely when  $W$  annihilates  $\ker(J)$ , and  $\mathcal{I}_W(J)$  is the trace of  $W$  against the inverse of  $J$  on  $\text{ran}(J)$ . For each  $t > 0$ , the map  $J \mapsto \text{Tr}[W(J + tI_p)^{-1}]$  is continuous on  $\mathbb{S}_+^p$ , and these functions increase pointwise as  $t \downarrow 0$ . Hence  $\mathcal{I}_W$  is lower semicontinuous.

We can now define the exact efficiency constant.

**DEFINITION 2.5 (Privacy constant).** Define

$$(11) \quad \mathfrak{E}_\theta(W) = \frac{1}{2} \inf_{J \in \mathcal{J}_\theta} \mathcal{I}_W(J).$$

By the compactness of  $\mathcal{J}_\theta$  and the lower semicontinuity of  $\mathcal{I}_W$ , the infimum is attained. When  $I(\theta) \succ 0$ , it is finite because  $\mathcal{J}_\theta$  contains a positive definite matrix. The following theorem identifies  $\mathfrak{E}_{\theta_0}(W)$  as the exact efficiency bound for the private local experiment.

**THEOREM 2.6 (Privacy-dominated constant).** Fix  $W \succeq 0$ . Under [Assumptions 1 and 2](#), suppose  $I(\theta_0) \succ 0$ . Let  $(\rho_n)$  satisfy the privacy-dominated regime in (9). Then, for every  $R_n \rightarrow \infty$  with  $R_n/\tau_n \rightarrow 0$ ,

$$(12) \quad \lim_{n \rightarrow \infty} \tau_n^2 \inf_{\hat{\theta}: \rho_n\text{-zCDP}} \sup_{\theta \in \Theta_n} \mathbb{E}_\theta \|\hat{\theta} - \theta\|_W^2 = \mathfrak{E}_{\theta_0}(W),$$

where  $\Theta_n$  is defined in (10).

**2.4. Unified efficiency across privacy regimes.** The preceding theorem concerns the privacy-dominated regime, in which the sampling fluctuation is asymptotically negligible. In the following, we establish a general efficiency theorem that covers different asymptotic regimes and unifies the efficiency constant. Recall that the statistic error and privacy error are of scale  $n^{-1}$  and  $\tau_n^{-2}$  respectively. For any positive sequence  $(\rho_n)$ , define the information ratio  $\lambda_n$ , the positive local scale  $\sigma_n$ , and the sampling weight  $\beta_n$  by

$$(13) \quad \lambda_n := n\rho_n = \frac{\tau_n^2}{n}, \quad \sigma_n^2 := \frac{1}{n} + \frac{1}{\tau_n^2}, \quad \beta_n := \frac{\lambda_n}{1 + \lambda_n}.$$

Then,  $\sigma_n^{-2}$  is the effective information obtained by reciprocal addition of the sampling information  $n$  and the privacy information  $\tau_n^2 = n^2 \rho_n$ . The weights  $\beta_n$  and  $1 - \beta_n$  are respectively the sampling and privacy shares of the common scale  $\sigma_n^2$ .

For a statistic  $\Phi : \mathcal{X} \rightarrow \mathbb{R}^d$ , write  $V_{\Phi, \theta} = \text{Var}_\theta(\Phi(X))$ . For  $\beta \in [0, 1]$ , define

$$(14) \quad \Sigma_{\Phi, \theta, \beta} = \beta V_{\Phi, \theta} + \frac{1 - \beta}{2} I_d.$$

When  $\beta < 1$ , this matrix is positive definite. The normalization in (13) is chosen so that the sampling and Gaussian privacy covariances satisfy the exact identity  $\sigma_n^2 \Sigma_{\Phi, \theta, \beta_n} = V_{\Phi, \theta} / n + I_d / (2\tau_n^2)$ .

DEFINITION 2.7 (Unified information region and constant). For  $\theta \in \Theta$  and  $\beta \in [0, 1]$ , let

$$(15) \quad \mathcal{U}_{\theta, \beta} = \text{cl} \left\{ A_{\Phi, \theta}^* \Sigma_{\Phi, \theta, \beta}^\dagger A_{\Phi, \theta} : d < \infty, \Phi : \mathcal{X} \rightarrow \mathbb{R}^d, \text{diam}(\Phi) \leq 1 \right\}.$$

The closure is taken in  $\mathbb{S}^p$ . For  $W \succeq 0$ , define

$$(16) \quad \mathfrak{C}_\theta^{(\beta)}(W) = \inf_{H \in \mathcal{U}_{\theta, \beta}} \mathcal{I}_W(H).$$

In the privacy and sampling regimes, respectively, the unified region recovers the private information region and Fisher information:

$$(17) \quad \mathcal{U}_{\theta, 0} = 2\mathcal{J}_\theta, \quad \mathfrak{C}_\theta^{(0)}(W) = \mathfrak{C}_\theta(W), \quad \mathfrak{C}_\theta^{(1)}(W) = \text{Tr}(WI(\theta)^{-1}).$$

For  $0 < \beta < 1$ , the covariance  $\Sigma_{\Phi, \theta, \beta}$  keeps the sampling and privacy contributions attached to the same statistic, so the constant remains a joint variational problem.

Define the unified local parameter set  $\tilde{\Theta}_n$  for a sequence  $(R_n)$  with

$$(18) \quad \tilde{\Theta}_n = \{\theta_0 + \sigma_n u : \|u\|_2 \leq R_n\}, \quad R_n \rightarrow \infty, \quad R_n \sigma_n \rightarrow 0.$$

THEOREM 2.8 (Unified efficiency). Fix  $W \succeq 0$ . Under [Assumptions 1 and 2](#), suppose  $I(\theta_0) \succ 0$ . Let  $(\rho_n)$  be a positive sequence such that

$$(19) \quad n^2 \rho_n \rightarrow \infty, \quad \beta_n \rightarrow \beta \in [0, 1],$$

where  $\beta_n$  is defined in (13). With  $\tilde{\Theta}_n$  defined in (18), we have

$$(20) \quad \lim_{n \rightarrow \infty} \sigma_n^{-2} \inf_{\hat{\theta} : \rho_n \text{-zCDP}} \sup_{\theta \in \tilde{\Theta}_n} \mathbb{E}_\theta \|\hat{\theta} - \theta\|_W^2 = \mathfrak{C}_{\theta_0}^{(\beta)}(W).$$

In particular, if the privacy constraint is weak, then we can achieve the optimal statistical error for free.

COROLLARY 2.9 (Sampling-dominated regime). Under the conditions of [Theorem 2.8](#), suppose that  $n\rho_n \rightarrow \infty$ . Then

$$(21) \quad \lim_{n \rightarrow \infty} n \inf_{\hat{\theta} : \rho_n \text{-zCDP}} \sup_{\theta \in \tilde{\Theta}_n} \mathbb{E}_\theta \|\hat{\theta} - \theta\|_W^2 = \text{Tr}(WI(\theta_0)^{-1}).$$

For the intermediate case when the privacy error is of the same order as the statistical error, the optimal constant is characterized by the variational constant (16) that reflects the delicate combination of two error sources, which is not a simple weighted average in general.

COROLLARY 2.10 (Transition regime). *Under the conditions of Theorem 2.8, suppose  $n\rho_n \rightarrow \lambda \in (0, \infty)$ , and put  $\beta = \lambda/(1 + \lambda)$ . Then*

$$(22) \quad \lim_{n \rightarrow \infty} n \inf_{\hat{\theta}: \rho_n\text{-zCDP}} \sup_{\theta \in \tilde{\Theta}_n} \mathbb{E}_\theta \|\hat{\theta} - \theta\|_W^2 = \frac{1}{\beta} \mathfrak{C}_{\theta_0}^{(\beta)}(W).$$

Finally, under strong privacy, the minimax quadratic risk over a fixed parameter set with nonempty interior remains bounded away from zero.

PROPOSITION 2.11 (Impossibility under strong privacy). *Under Assumptions 1 and 2, fix  $W \succeq 0$  with  $W \neq 0$ , and suppose  $\Theta$  has nonempty interior. If  $n^2\rho_n = O(1)$ , then there exists a constant  $c > 0$  such that for all sufficiently large  $n$ ,*

$$(23) \quad \inf_{\hat{\theta}: \rho_n\text{-zCDP}} \sup_{\theta \in \Theta} \mathbb{E}_\theta \|\hat{\theta} - \theta\|_W^2 \geq c.$$

2.5. *Unified efficiency on compact parameter sets.* The preceding local theorem treats the center  $\theta_0$  as public and known. On a fixed compact parameter set, the same local constant can be attained without a public center by first releasing a private preliminary estimator and then selecting one of finitely many local Gaussian constructions. Because  $\mathfrak{C}_\theta^{(\beta)}(W)$  may vary with  $\theta$ , our uniform statement normalizes the risk pointwise by this local constant.

THEOREM 2.12 (Unified efficiency on a compact parameter set). *Fix  $W \succeq 0$  with  $W \neq 0$ , and let  $K \subset \Theta$  be a nonempty compact set satisfying  $K = \overline{K^\circ}$ . Under Assumptions 1 and 2, suppose  $I(\theta) \succ 0$  for every  $\theta \in K$  and  $\theta \mapsto I(\theta)$  is continuous on  $K$ . Assume the model is identifiable on  $K$ , in the sense that  $P_{\theta_1} \neq P_{\theta_2}$  whenever  $\theta_1, \theta_2 \in K$  and  $\theta_1 \neq \theta_2$ . Let  $(\rho_n)$  be a positive sequence satisfying the asymptotic regime in (19), where  $\beta_n$  is defined in (13). Then  $\theta \mapsto \mathfrak{C}_\theta^{(\beta)}(W)$  is continuous and takes values in  $(0, \infty)$  on  $K$ , and*

$$(24) \quad \lim_{n \rightarrow \infty} \inf_{\hat{\theta}: \rho_n\text{-zCDP}} \sup_{\theta \in K} \frac{\sigma_n^{-2} \mathbb{E}_\theta \|\hat{\theta} - \theta\|_W^2}{\mathfrak{C}_\theta^{(\beta)}(W)} = 1.$$

COROLLARY 2.13. *Under the assumptions of Theorem 2.12,*

$$\lim_{n \rightarrow \infty} \sigma_n^{-2} \inf_{\hat{\theta}: \rho_n\text{-zCDP}} \sup_{\theta \in K} \mathbb{E}_\theta \|\hat{\theta} - \theta\|_W^2 = \max_{\theta \in K} \mathfrak{C}_\theta^{(\beta)}(W).$$

On the other hand, if the parameter space is unbounded, then the minimax lower bound is infinite under privacy.

PROPOSITION 2.14 (Impossibility for unbounded parameter). *Fix  $W \succeq 0$ , and suppose  $\sup_{\theta, \vartheta \in \Theta} \|\theta - \vartheta\|_W^2 = \infty$ . Then, for every  $n \geq 1$  and every finite  $\rho \geq 0$ ,*

$$\inf_{\hat{\theta}: \rho\text{-zCDP}} \sup_{\theta \in \Theta} \mathbb{E}_\theta \|\hat{\theta} - \theta\|_W^2 = \infty.$$

2.6. *Discussion.* We conclude this section with several implications of the information region and the unified local and compact efficiency theorems.

2.6.1. *Privacy regimes and optimal constants.* Theorems 2.6, 2.8 and 2.12 identify the exact minimax constant for private estimation under zCDP in their respective local and compact settings. For every quadratic loss with  $W \succeq 0$ , the privacy-dominated local theorem shows that the risk, normalized by  $\tau_n^2$ , converges to the variational constant  $\mathfrak{C}_{\theta_0}(W)$ . The unified local theorem replaces  $\tau_n^2$  by  $\sigma_n^{-2}$  and connects the privacy-dominated, transition, and Fisher constants, while the compact theorem attains the corresponding regime-dependent local constant uniformly over a compact parameter set. Because the minimization ranges over all  $\rho_n$ -zCDP mechanisms and all estimators based on their outputs, these results characterize a fundamental limit rather than the performance of a particular mechanism class.

The two compound scales in the theory play different roles (Table 1). The privacy-induced information scale is  $\tau_n^2 = n^2 \rho_n$  and the ratio  $\lambda_n = n \rho_n = \tau_n^2/n$  compares privacy and sampling information and determines their first-order balance. Under strong privacy  $\tau_n^2 = O(1)$ , Proposition 2.11 gives a nonvanishing lower bound. When  $\tau_n^2 \rightarrow \infty$  and  $\lambda_n \rightarrow 0$  ( $\beta = 0$ ), estimation is consistent but privacy dominated. When  $\lambda_n \rightarrow \infty$  ( $\beta = 1$ ), privacy is negligible on the classical root- $n$  scale and Fisher efficiency is recovered. When  $\lambda_n \rightarrow \lambda \in (0, \infty)$  ( $\beta \in (0, 1)$ ), sampling and privacy contribute on the same scale, and their exact constant is a joint variational quantity.

Our results apply broadly across regular fixed-dimensional parametric models. Nevertheless, the optimal variational constant  $\mathfrak{C}_{\theta}^{(\beta)}(W)$  may be difficult to compute in general because characterizing the region  $\mathcal{U}_{\theta, \beta}$  requires optimization over bounded statistics. In Section 5, we investigate several concrete models and give closed-form expressions for  $\mathfrak{C}_{\theta}^{(\beta)}(W)$ .

2.6.2. *Set-valued private information geometry.* The variational expression for  $\mathfrak{C}_{\theta}(W)$  also explains why the relevant private information object is a region rather than a single matrix. Privacy does not simply reduce Fisher information uniformly across directions, but it creates trade-offs among the information attainable in different directions. Indeed, whereas classical Fisher information describes all local directions through the single quadratic form  $h \mapsto h^\top I(\theta) h$ , Proposition 2.4 identifies  $\sup_{J \in \mathcal{J}_{\theta}} h^\top J h$  with the squared local total variation speed. The squared directional speed need not be quadratic in  $h$ , so no single feasible matrix need attain the supremum in every direction. A maximizing score-sign statistic may consequently depend on  $h$ , and statistics that are separately optimal in different directions need not be compatible under a common diameter constraint. The region  $\mathcal{J}_{\theta}$  records precisely this compatibility by specifying which combinations of directional information can be attained jointly.

Our private information geometry is closely related to the  $L^1$ -information studied by Duchi and Ruan (2024) in local DP estimation. Along the one-dimensional submodel  $t \mapsto \theta + th$ , their  $L^1$ -information is  $\mathbb{E}_{\theta} |\langle s_{\theta}(X), h \rangle|$ , which is precisely the directional total variation speed in (7). In particular, when  $p = 1$ , the region reduces to the interval in Example 2.1. The distinction in several dimensions is that a local DP benchmark may treat directions separately, whereas our central zCDP problem requires a single mechanism to operate uniformly over a multidimensional local parameter set. The region  $\mathcal{J}_{\theta}$  records which directional information levels are jointly attainable and thereby yields the exact minimax constant under a general quadratic loss.

The structural properties in Proposition 2.3 give this geometry an operational interpretation. Combining feasible statistics in orthogonal components yields convexity, while discarding attainable information yields downward closure. Consequently, only the upper Pareto boundary of  $\mathcal{J}_{\theta}$  can be relevant for efficiency. Fisher information remains a useful outer envelope: (8) gives  $J \preceq I(\theta)/4$  for every  $J \in \mathcal{J}_{\theta}$ , and  $I(\theta) \succ 0$  is equivalent to the existence of a positive definite feasible information matrix. It determines whether nondegenerate private information is available and supplies a universal quadratic bound, but it does not generally

determine the exact private efficiency constant. In this sense, privacy changes the local information geometry rather than merely reducing the effective sample size.

The same set-valued interpretation extends to the unified information region  $\mathcal{U}_{\theta,\beta}$ . Its elements record the information matrices jointly attainable when the same bounded statistic determines both the score–statistic covariance and the sampling covariance. In the privacy and sampling regimes,  $\mathcal{U}_{\theta,0} = 2\mathcal{J}_\theta$ ,  $\mathfrak{C}_\theta^{(0)}(W) = \mathfrak{C}_\theta(W)$ , and  $\mathfrak{C}_\theta^{(1)}(W) = \text{Tr}(WI(\theta)^{-1})$ , while for  $0 < \beta < 1$  the region retains both privacy and sampling contributions. Thus, as the sampling weight changes, the unified theory changes the attainable information geometry itself rather than merely rescaling a fixed private region.

**2.6.3. Loss-dependent efficiency.** The efficiency constant in (16) is given by an optimization problem depending on  $W$ , and the minimizer (if it exists) need not be the same for every  $W$ , particularly when  $\beta \in [0, 1)$ . Consequently, there need not be a single statistic, and hence a single private estimator, that is always efficient. This is in sharp contrast to the classical theory about efficiency, where a single MLE is efficient for every  $W$ . The following proposition formalizes this loss dependency and incompatibility.

**PROPOSITION 2.15 (Loss-specific incompatibility).** Suppose  $I(\theta_0) \succ 0$ , fix  $\beta \in [0, 1]$ , and let  $W_1, W_2 \succeq 0$ . Write  $\mathcal{A}_{\theta_0,\beta}(W) = \arg \min_{H \in \mathcal{U}_{\theta_0,\beta}} \mathcal{I}_W(H)$ . Then

$$(25) \quad \mathfrak{C}_{\theta_0}^{(\beta)}(W_1 + W_2) \geq \mathfrak{C}_{\theta_0}^{(\beta)}(W_1) + \mathfrak{C}_{\theta_0}^{(\beta)}(W_2),$$

with equality if and only if  $\mathcal{A}_{\theta_0,\beta}(W_1) \cap \mathcal{A}_{\theta_0,\beta}(W_2) \neq \emptyset$ . If these optimizer sets are disjoint, no single sequence of  $\rho_n$ -zCDP estimators can attain both local asymptotic minimax constants  $\mathfrak{C}_{\theta_0}^{(\beta)}(W_1)$  and  $\mathfrak{C}_{\theta_0}^{(\beta)}(W_2)$  along any sequence of privacy budgets satisfying (19).

**2.6.4. Attainability and universality.** The variational problem directly determines asymptotically attaining procedures. For a fixed  $\beta$ , a statistic  $\Phi$  of diameter one generates the unified information matrix  $H_{\Phi,\theta_0,\beta} = A_{\Phi,\theta_0}^* \Sigma_{\Phi,\theta_0,\beta}^\dagger A_{\Phi,\theta_0}$ . Releasing its empirical mean with Gaussian noise and applying projected local inversion yields the inverse-information risk associated with this matrix. The blockwise construction in Theorems 3.4 and 3.8 therefore turns this variational approximation into a single asymptotically efficient estimator sequence in the local and compact settings. For the transition and Fisher regimes, the proofs use the same projected inversion with the sampling covariance retained in the local risk calculation. Theorem 3.5 gives a local attaining sequence with uniform Gaussian limit  $N(0, H_{W,\beta}^{-1})$  for  $W \succ 0$ , where  $H_{W,\beta}$  denotes the unique optimizing information matrix.

Our minimax characterization establishes the first-order universality of Gaussian mechanisms for efficiency problems considered here. Across different models, the exact minimax constant is attained by a Gaussian release of a suitably chosen bounded statistic, followed by moment-map inversion. Given the minimax lower bound, it is unnecessary to resort to other nonadditive, nonlinear, or otherwise unrestricted private mechanisms for improvement. The essential choice is therefore not a more elaborate noise distribution, but the statistic to which Gaussian noise is added.

**2.6.5. From a public center to uniform efficiency.** The local theorem treats  $\theta_0$  as public because the optimal statistic and its inverse mean map may depend on the center. Under the assumptions of Theorem 2.12, including global identifiability on  $K$ , the uniform theorem removes this requirement without first-order loss. The bounded statistic  $T$  serves only to localize the unknown parameter, whereas the statistics used in the second stage approximate the loss-specific information optimum on finitely many local charts. The statistic used to identify the chart therefore need not itself be efficient for final estimation.

The first-stage localization uses asymptotically negligible shares of the sample and privacy budget, so the second stage preserves the pointwise leading constant  $\mathfrak{C}_\theta^{(\beta)}(W)$  uniformly over  $K$ . The normalized theorem states this pointwise efficiency, while [Corollary 2.13](#) shows that the unnormalized compact minimax risk is governed by the least favorable local constant  $\max_{\theta \in K} \mathfrak{C}_\theta^{(\beta)}(W)$ .

**3. Optimal Estimator via Gaussian Release and Moment-Map Inversion.** This section develops the Gaussian mechanisms that attain the upper bounds in [Theorems 2.8](#) and [2.12](#). We first construct a moment-map inverse estimator from a fixed bounded statistic, and then choose the statistic to approach the optimal variational constant. Finally, we use a private preliminary estimator to remove the assumption that the local center is publicly known. We first recall two standard properties of  $\rho$ -zCDP ([Dwork and Rothblum, 2016](#); [Bun and Steinke, 2016](#)).

LEMMA 3.1. *We have the following properties of  $\rho$ -zCDP.*

- Gaussian mechanism. *If  $f : \mathcal{X}^n \rightarrow \mathbb{R}^d$  has replacement  $\ell_2$ -sensitivity*

$$\Delta = \sup_{S, S' \text{ adjacent}} \|f(S) - f(S')\|_2,$$

*then  $f(S) + Z$ , where  $Z \sim N(0, \Delta^2 I_d / (2\rho))$ , is  $\rho$ -zCDP.*

- Adaptive composition. *If, conditional on any preceding outputs, the  $j$ th mechanism is  $\rho_j$ -zCDP for  $j = 1, \dots, k$ , then their adaptive composition is  $(\sum_{j=1}^k \rho_j)$ -zCDP.*

3.1. *Local upper bound for a bounded statistic.* First, suppose we have a bounded statistic with a nonsingular mean derivative. As introduced in [Section 1](#), this gives a private estimator by releasing its empirical mean and locally inverting its population moment map. We recall the construction. Fix  $\theta_0 \in \Theta$ , and let  $\Phi : \mathcal{X} \rightarrow \mathbb{R}^p$  satisfy  $\text{diam}(\Phi) \leq 1$ . Suppose its mean map  $g(\theta) = \mathbb{E}_\theta \Phi(X)$  is continuously differentiable near  $\theta_0$ , with nonsingular derivative  $A_0 = Dg(\theta_0) = A_{\Phi, \theta_0}$ . Let  $\Psi_0$  be the local  $C^1$  inverse of  $g$  on a small closed Euclidean ball  $\mathcal{B}_0$  centered at  $g(\theta_0)$ , and let  $\Pi_{\mathcal{B}_0}$  denote Euclidean projection onto this ball. Set  $F_0 = \Psi_0 \circ \Pi_{\mathcal{B}_0}$ . Define the Gaussian release and the corresponding moment-map inverse estimator by

$$(26) \quad \hat{\theta}_{\Phi, n} = F_0(Y_n), \quad Y_n = \frac{1}{n} \sum_{i=1}^n \Phi(X_i) + G_n, \quad G_n \sim N\left(0, \frac{I_p}{2\tau_n^2}\right).$$

By [Lemma 3.1](#), the estimator is  $\rho_n$ -zCDP: the empirical mean has replacement sensitivity at most  $1/n$ , and the remaining operations are post-processing. Its sampling covariance and Gaussian privacy covariance satisfy

$$\text{Var}_\theta(Y_n - g(\theta)) = \frac{V_{\Phi, \theta}}{n} + \frac{I_p}{2\tau_n^2} = \sigma_n^2 \Sigma_{\Phi, \theta, \beta_n}.$$

Thus the covariance appearing in [\(14\)](#) is exactly the first-order covariance of the released moment on the  $\sigma_n$ -scale. The next theorem identifies the resulting local risk.

THEOREM 3.2 (Local upper bound). *Under [Assumptions 1](#) and [2](#), suppose the statistic above has the stated continuously differentiable mean map and nonsingular derivative. Let  $(\rho_n)$  satisfy the asymptotic regime in [\(19\)](#). For every  $(R_n)$  and  $\tilde{\Theta}_n$  satisfying [\(18\)](#),  $\hat{\theta}_{\Phi, n}$  is  $\rho_n$ -zCDP and, for every  $W \succeq 0$ ,*

$$(27) \quad \sup_{\theta \in \tilde{\Theta}_n} \left| \sigma_n^{-2} \mathbb{E}_\theta \|\hat{\theta}_{\Phi, n} - \theta\|_W^2 - \text{Tr} \left[ W A_0^{-1} \Sigma_{\Phi, \theta_0, \beta} A_0^{-\top} \right] \right| \longrightarrow 0.$$

The idea of [Theorem 3.2](#) is simple. After the local inverse transformation, the released moment is adjusted by the derivative  $A_0^{-1}$  of the inverse map. Consequently, the sampling and privacy fluctuations have first-order covariance  $A_0^{-1}\Sigma_{\Phi,\theta_0,\beta}A_0^{-\top}$ . The projection makes the estimator globally defined without affecting this local expansion. The details are given in Section S2 of the Supplementary Material ([Cai and Li, 2026a](#)).

**3.2. Near-optimal statistics and local attainment.** Next, we show that the bound in [Theorem 3.2](#) can be made arbitrarily close to  $\mathfrak{C}_{\theta_0}^{(\beta)}(W)$  by choosing a nearly optimal statistic.

**LEMMA 3.3** (Nearly optimal statistic). *Fix  $W \succeq 0$ ,  $\theta_0 \in \Theta$ , and  $\beta \in [0, 1]$ . Under [Assumptions 1 and 2](#), suppose  $I(\theta_0) \succ 0$ . For every  $\epsilon > 0$ , there is a statistic  $\Phi_\epsilon : \mathcal{X} \rightarrow \mathbb{R}^p$ , with  $\text{diam}(\Phi_\epsilon) \leq 1$ , whose mean map is continuously differentiable near  $\theta_0$ , whose derivative  $A_\epsilon = A_{\Phi_\epsilon, \theta_0}$  is nonsingular, and such that*

$$(28) \quad \text{Tr} \left[ W A_\epsilon^{-1} \Sigma_{\Phi_\epsilon, \theta_0, \beta} A_\epsilon^{-\top} \right] \leq \mathfrak{C}_{\theta_0}^{(\beta)}(W) + \epsilon.$$

For  $\beta < 1$ , the argument perturbs an optimizing information matrix toward a positive-definite feasible matrix, approximates the result by an actual diameter-one statistic, and reduces that statistic to  $\mathbb{R}^p$ . At  $\beta = 1$ , the same conclusion follows from bounded truncated-score statistics approaching the Fisher information limit. The assumed  $L^1$  differentiability and the inverse function theorem then provide the local inverse required in [Theorem 3.2](#). Details are given in Section S2 of the Supplementary Material ([Cai and Li, 2026a](#)).

Combining [Lemma 3.3](#) with [Theorem 3.2](#) and taking a diagonal sequence yields the following result, which is the constructive upper half of [Theorem 2.8](#) under the same assumptions.

**THEOREM 3.4** (Local optimal estimator). *Fix  $W \succeq 0$ . Under [Assumptions 1 and 2](#), suppose  $I(\theta_0) \succ 0$ . Let  $(\rho_n)$  satisfy the asymptotic regime in (19). Then, for every  $(R_n)$  and  $\tilde{\Theta}_n$  satisfying (18), there is a sequence  $(\hat{\theta}_n^*)$  of  $\rho_n$ -zCDP moment-map inverse estimators, each based on an  $\mathbb{R}^p$ -valued statistic of diameter at most one, such that*

$$(29) \quad \limsup_{n \rightarrow \infty} \sigma_n^{-2} \sup_{\theta \in \tilde{\Theta}_n} \mathbb{E}_\theta \|\hat{\theta}_n^* - \theta\|_W^2 \leq \mathfrak{C}_{\theta_0}^{(\beta)}(W).$$

The preceding risk bounds admit a distributional refinement: the optimal sequence can be chosen to have a uniform Gaussian limit.

**THEOREM 3.5** (Gaussian limit for attaining sequences). *Under the assumptions of [Theorem 3.4](#), suppose  $W \succ 0$  and take the unique minimizer  $H_{W,\beta} = \arg \min_{H \in \mathcal{U}_{\theta_0,\beta}} \mathcal{I}_W(H)$ . The attaining sequence may be chosen so that*

$$(30) \quad \sigma_n^{-1}(\hat{\theta}_n^* - \theta) \rightsquigarrow N\left(0, H_{W,\beta}^{-1}\right) \quad \text{uniformly over } \theta \in \tilde{\Theta}_n.$$

This result parallels the asymptotic normality of efficient estimators in classical regular models, where the local limit has covariance  $I(\theta_0)^{-1}$  at the  $n^{-1/2}$  scale ([van der Vaart, 1998](#), Chapter 8). In the sampling-dominated regime  $\beta = 1$ , we have  $\sigma_n^{-1} \sim \sqrt{n}$  and  $H_{W,1} = I(\theta_0)$ , so (30) recovers this classical limit. For  $0 \leq \beta < 1$ , privacy changes both the effective scale and the attainable covariance: the optimizing information matrix  $H_{W,\beta}$  is selected from  $\mathcal{U}_{\theta_0,\beta}$  and may depend on the loss  $W$ . For each fixed  $W \succ 0$ , the optimizing information matrix is unique: when  $\beta < 1$ , the region  $\mathcal{U}_{\theta_0,\beta}$  is convex and  $H \mapsto \text{Tr}(WH^{-1})$  is strictly convex on the positive definite cone, while at  $\beta = 1$  the unique optimizer is  $I(\theta_0)$ . Different losses may therefore select different unique optimizing matrices, although statistics generating the same matrix need not be unique.

3.3. *Preliminary localization on compact parameter sets.* The local constructions above require the public center  $\theta_0$  to be known. In this subsection, we show that a private preliminary estimator can select both the local center and the local statistic while using only vanishing fractions of the sample and privacy budget.

The following lemma shows that, under the remaining regularity and nonsingularity conditions in [Theorem 2.12](#), global identifiability supplies the bounded finite-dimensional moment identifier needed for preliminary localization.

LEMMA 3.6 (Bounded identifier). *Let  $K \subset \Theta$  be nonempty and compact. Under [Assumptions 1](#) and [2](#), suppose  $I(\theta) \succ 0$  for every  $\theta \in K$ , and suppose the model is identifiable on  $K$ . Then there are a finite integer  $m$  and a bounded measurable statistic  $T : \mathcal{X} \rightarrow \mathbb{R}^m$ , which may be chosen with  $\text{diam}(T) \leq 1$ , such that*

$$h(\theta) = \mathbb{E}_\theta T(X)$$

*is continuously differentiable on  $\Theta$  and one-to-one on  $K$ .*

Let  $K \subset \Theta$  be nonempty and compact, and let  $T : \mathcal{X} \rightarrow \mathbb{R}^m$  have finite diameter  $\Delta_T = \text{diam}(T)$ , and suppose its mean map  $h$  is continuous and one-to-one on  $K$ . Under the assumptions of [Theorem 2.12](#), such a statistic is supplied by [Lemma 3.6](#). For  $n_1$  observations and privacy budget  $\rho_1$ , define the preliminary localization estimator  $\tilde{\theta}$  by

$$(31) \quad Y_1 = \frac{1}{n_1} \sum_{i=1}^{n_1} T(X_i) + G_{1,n}, \quad G_{1,n} \sim N\left(0, \frac{\Delta_T^2 I_m}{2n_1^2 \rho_1}\right),$$

$$\tilde{\theta} \in \arg \min_{\vartheta \in K} \|Y_1 - h(\vartheta)\|_2.$$

The next lemma shows that this estimator attains the error probability needed by the second stage while using only a small share of the available resources.

LEMMA 3.7 (Private preliminary localization). *The estimator in (31) is  $\rho_1$ -zCDP. Let  $a_n \rightarrow \infty$ . If  $n_1 / \log a_n \rightarrow \infty$  and  $n_1^2 \rho_1 / \log a_n \rightarrow \infty$ , then, for every fixed  $r > 0$ ,*

$$\sup_{\theta \in K} P_\theta \left\{ \|\tilde{\theta} - \theta\|_2 > r \right\} = o(a_n^{-2}).$$

To pass from preliminary localization to compact attainment, take  $a_n = \sigma_n^{-1}$ . Compactness yields finitely many local charts, each equipped with a diameter-one statistic whose Gaussian release and projected local inverse are nearly optimal for  $\mathfrak{C}_\theta^{(\beta)}(W)$  throughout the chart. We spend a vanishing fraction of the sample and privacy budget on the preliminary estimator, use it to select a valid chart, and apply the corresponding Gaussian mechanism to the remaining observations. Adaptive composition preserves the total privacy budget, while uniform localization and the negligible first stage leave the leading risk unchanged on the  $\sigma_n^2$ -scale.

THEOREM 3.8 (Uniform optimal estimator). *Under the assumptions of [Theorem 2.12](#), there is a sequence  $(\hat{\theta}_n^*)$  of  $\rho_n$ -zCDP estimators, eventually obtained by the two-stage Gaussian construction described above, such that*

$$\limsup_{n \rightarrow \infty} \sup_{\theta \in K} \frac{\sigma_n^{-2} \mathbb{E}_\theta \|\hat{\theta}_n^* - \theta\|_W^2}{\mathfrak{C}_\theta^{(\beta)}(W)} \leq 1.$$

**4. Lower Bounds over All zCDP Mechanisms.** In this section, we establish optimality through minimax lower bounds over all zCDP procedures. For a mechanism  $M$ , let  $Q_\theta^M = M \circ P_\theta^{\otimes n}$  denote the output distribution, and let  $I_M(\theta)$  denote the Fisher information matrix of the output experiment. Restricting attention to standard Borel outputs entails no loss, since composing any transcript with its final  $\mathbb{R}^p$ -valued estimator preserves zCDP by post-processing.

**4.1. Information-region contraction.** The main idea for the lower bound is to show that an arbitrary private mechanism cannot carry more local information than the information region in [Definition 2.2](#). We have the following theorem.

**THEOREM 4.1** (Information-region contraction under zCDP). *Suppose [Assumption 1](#) holds. Let  $M$  be an arbitrary  $\rho$ -zCDP Markov kernel from  $\mathcal{X}^n$  into a standard Borel output space  $\mathcal{Z}$ . Then there is a scalar  $\delta_\rho = 1 + o_\rho(1)$  as  $\rho \downarrow 0$ , depending only on  $\rho$ , such that for every such  $M$  and every  $\theta \in \Theta$ , there exists  $J_{M,\theta,\rho} \in \mathcal{J}_\theta$  with*

$$(32) \quad I_M(\theta) = 2\rho n^2 \delta_\rho^2 J_{M,\theta,\rho}.$$

*The  $o_\rho(1)$  term is uniform over mechanisms and over parameter values for which the assumptions hold.*

Taking the trace and using [\(8\)](#) yields the following corollary.

**COROLLARY 4.2.** *Under the same settings as in [Theorem 4.1](#), we have*

$$(33) \quad \text{Tr } I_M(\theta) \leq 2\rho n^2 \Gamma_\theta(1 + o_\rho(1)) \leq \frac{1}{2} \rho n^2 \text{Tr } I(\theta)(1 + o_\rho(1)).$$

[Theorem 4.1](#) shows that the Fisher information of a  $\rho$ -zCDP mechanism is asymptotically restricted to  $\mathcal{J}_\theta$  after normalization. In comparison to the rough upper bound in the private information contraction lemma in [Cai and Li \(2026b\)](#), [Theorem 4.1](#) is more refined to capture the constant dependency. In particular, the last inequality in [Corollary 4.2](#) recovers their private Fisher information contraction. Moreover, the matrix representation is stronger than its trace consequence and retains the anisotropic geometry needed for general quadratic loss.

To prove [Theorem 4.1](#), the fingerprinting attack or score attack ([Cai, Wang and Zhang, 2023](#)) commonly used in the literature no longer works, as they are too coarse for our purpose. We develop the following lemma to transfer  $\rho$ -zCDP condition to the boundedness of the likelihood ratio as  $L^2$ -valued statistics.

**LEMMA 4.3** (Barycentric likelihood ratio diameter). *Let  $\{R_t : t \in T\}$  be a probability kernel and let  $\lambda$  be a probability measure on  $T$ . Suppose the likelihood ratios admit jointly measurable versions and  $D_\alpha(R_t \| R_s) \leq \alpha\rho$  for all  $s, t \in T$  and  $\alpha > 1$ . If  $\bar{R} = \int R_t \lambda(dt)$  and  $r_t = dR_t/d\bar{R}$ , then, uniformly over the family and mixing measure,*

$$(34) \quad \sup_{s,t \in T} \|r_t - r_s\|_{L^2(\bar{R})}^2 \leq 2\rho(1 + o_\rho(1)).$$

For each record  $i$ , let  $Q_{i,x}$  be the output law obtained by fixing that record at  $x$  and averaging over the remaining observations. Then  $Q_\theta^M = \int Q_{i,x} P_\theta(dx)$  is the barycenter of this conditional output family. Joint convexity transfers the zCDP bounds for neighboring deterministic data sets to pairwise Rényi bounds for  $Q_{i,x}$  and  $Q_{i,x'}$ . Thus [Lemma 4.3](#), applied with  $L_{i,x} = dQ_{i,x}/dQ_\theta^M$ , shows that the Hilbert-valued statistic  $\Phi_i(x) = L_{i,x}/(\sqrt{2\rho\delta_\rho})$  has diameter at most one in  $L^2(Q_\theta^M)$ , so is their average  $\bar{\Phi}(x)$ . Then, a finite dimensional compression

yields an at most  $p$ -dimensional statistic with no larger diameter and the same associated information matrix  $A_{\bar{\Phi},\theta}^* A_{\bar{\Phi},\theta}$ , so this matrix must be contained in  $\mathcal{J}_\theta$  by its definition. On the other hand, we can establish identity between the Fisher information matrix  $I_M(\theta)$  and the information matrix of  $L_{i,x}$ . Plugging in the scaling concludes [Theorem 4.1](#). The full proof is given in Section S4.4 of the Supplementary Material ([Cai and Li, 2026a](#)).

**4.2. Local minimax lower bound under privacy regime.** With the local information contraction, we can use a van Trees argument to obtain the local minimax lower bound.

**THEOREM 4.4 (Minimax lower bound).** *Fix  $\theta_0 \in \Theta$  and  $W \succeq 0$ . Under [Assumptions 1 and 2](#), let  $(\rho_n)$  satisfy the privacy regime in (9). Then, with  $\Theta_n$  defined in (10),*

$$\liminf_{n \rightarrow \infty} \tau_n^2 \inf_{\hat{\theta}: \rho_n\text{-zCDP}} \sup_{\theta \in \Theta_n} \mathbb{E}_\theta \|\hat{\theta} - \theta\|_W^2 \geq \mathfrak{C}_{\theta_0}(W).$$

The proof of [Theorem 4.4](#) is given in Section S4.7 of the Supplementary Material ([Cai and Li, 2026a](#)). Together with the local optimal estimator in [Theorem 3.4](#), [Theorem 4.4](#) completes the proof of the local efficiency theorem. The same lower bound, applied inside compact parameter sets and combined with [Theorem 3.8](#), proves the compact efficiency theorem in the privacy-dominated regime.

**4.3. Transition regime.** The privacy contraction above fails to give the sharp constant in the transition regime where the sampling fluctuation is not negligible. A more refined analysis is needed to characterize the sampling and privacy contributions jointly for every mechanism.

To this end, we first consider the  $k$ -category model, for which the information can be computed more explicitly. A key observation is that symmetrization reduces an arbitrary mechanism to a channel  $m \mapsto Q_m$  from the multinomial count vector, without changing either its output law under iid sampling or its zCDP guarantee. The output information then satisfies an identity for missing information: the full multinomial information is reduced by a term determined by the conditional covariance of the counts given the mechanism output. To control this loss sharply, we compare the channel at neighboring count vectors  $m$  and  $m + e_a - e_k$ . The Gram matrix of the resulting shift scores separates, to first order, into a sampling contribution  $V_q^{-1}/n$  from the multinomial count law and a privacy contribution  $2\rho_n J_{n,q}$ , where  $V_q$  is the covariance matrix of the first  $d = k - 1$  category indicators and  $J_{n,q}$  is the Gram matrix of a configuration with pairwise distances at most one. A matrix projection inequality converts this decomposition into an upper bound on the output Fisher information. Consequently, the sampling and privacy contributions, though of the same order, can be decomposed separately through

$$(35) \quad \frac{1}{n} I_{M_n}(q) \leq V_q^{-1} - V_q^{-1} (V_q^{-1} + 2\lambda_n J_{n,q})^{-1} V_q^{-1} + o(1) I_d, \quad \lambda_n = n\rho_n.$$

For the general regular model, we freeze the conditional distribution within each cell of a finite measurable partition and vary only the cell probabilities. The  $L^1$ -smoothness assumption permits a partition whose cell score approximates the full score arbitrarily well. The bound on a finite alphabet then transfers uniformly over all mechanisms: for every  $\epsilon > 0$ , a shrinking prior  $\pi_n$  supported on the local parameter set satisfies, for a finite constant  $C_{\theta_0,\lambda}$  independent of the mechanism,

$$(36) \quad \frac{1}{n} \int I_{M_n}(\theta) \pi_n(d\theta) \leq L_{n,\epsilon} + (C_{\theta_0,\lambda}(\epsilon + \epsilon^2) + o(1)) I_p, \quad L_{n,\epsilon} \in \beta\mathcal{U}_{\theta_0,\beta}.$$

The prior can be chosen so that its own information is negligible on the root- $n$  scale, namely  $J_{\pi_n}/n \rightarrow 0$ . Applying the matrix van Trees inequality to the output experiment, then letting  $n \rightarrow \infty$  and  $\epsilon \downarrow 0$ , gives

$$(37) \quad \liminf_{n \rightarrow \infty} n \int \mathbb{E}_\theta \|\hat{\theta} - \theta\|_W^2 \pi_n(d\theta) \geq \inf_{L \in \beta \mathcal{U}_{\theta_0, \beta}} \mathcal{I}_W(L) = \frac{1}{\beta} \mathfrak{C}_{\theta_0}^{(\beta)}(W).$$

Since the Bayes risk is bounded above by the local worst-case risk, this proves the lower-bound half of [Corollary 2.10](#) for arbitrary mechanisms. The technical details are given in Section S5 of the Supplementary Material ([Cai and Li, 2026a](#)).

**5. Examples.** In this section, we give explicit expressions of the efficiency constants  $\mathfrak{C}_{\theta_0}^{(\beta)}(W)$  for various parametric models. For simplicity, we focus on local constructions with a public center  $\theta_0$ , while private localization is covered by [Theorem 2.12](#). For more examples and a summary, see Section S7 and S8 in the Supplementary Material ([Cai and Li, 2026a](#)).

**5.1. One-dimensional regular families.** Consider the one-dimensional model with scalar target

$$(38) \quad X_1, \dots, X_n \stackrel{\text{iid}}{\sim} P_\theta, \quad \theta \in \Theta \subset \mathbb{R}, \quad \mu : \Theta \rightarrow \mathbb{R}.$$

**PROPOSITION 5.1 (One-dimensional constant).** For the model and target in (38), suppose  $\mu$  is continuously differentiable, with  $\mu'(\theta_0) \neq 0$ . Under [Assumptions 1](#) and [2](#), suppose  $I(\theta_0) > 0$ . For  $\beta \in [0, 1]$ , define

$$(39) \quad \mathfrak{J}_{\theta_0}^{(\beta)} = \sup_{\substack{\phi: \mathcal{X} \rightarrow \mathbb{R} \\ \text{diam}(\phi) \leq 1}} \frac{(\mathbb{E}_{\theta_0} [\phi(X) s_{\theta_0}(X)])^2}{\beta \text{Var}_{\theta_0}(\phi(X)) + (1 - \beta)/2}.$$

Then, writing  $a_{\theta_0} = \mathbb{E}_{\theta_0} |s_{\theta_0}(X)|/2$ , the exact local minimax constant of  $\mu(\theta)$  is

$$(40) \quad \mathfrak{C}_{\mu, \theta_0}^{(\beta)} = \frac{(\mu'(\theta_0))^2}{\mathfrak{J}_{\theta_0}^{(\beta)}}, \quad \mathfrak{C}_{\mu, \theta_0}^{(0)} = \frac{(\mu'(\theta_0))^2}{2a_{\theta_0}^2}, \quad \mathfrak{C}_{\mu, \theta_0}^{(1)} = \frac{(\mu'(\theta_0))^2}{I(\theta_0)}.$$

The statistics for optimal estimators have simple forms in the privacy or sampling regime. In the privacy regime  $\beta = 0$ , the optimal statistic is the score sign statistic  $\phi_{\text{priv}}(x) = \frac{1}{2} \text{sign}(s_{\theta_0}(x))$ , as in [Example 2.1](#). In the sampling regime  $\beta = 1$ , the optimal statistic is given by  $\phi_{\text{stat}}(x) \propto s_{\theta_0}(x)$  with a proper rescaling and truncation approximation if the score is unbounded.

Moreover, we give explicit examples for the one-dimensional regular families. See [Table 2](#). Specifically, the logistic location model is given by distribution function  $F_\theta(x) = 1/(1 + \exp(-(x - \theta)/s))$ ,  $\theta \in \mathbb{R}$ , where the scale  $s > 0$  is known and the target is  $\mu(\theta) = \theta$ . Denote  $r_\beta = (1 - \sqrt{1 - \beta})/(1 + \sqrt{1 - \beta})$ . Both the Bernoulli and Laplace location models have binary scores: up to an affine transformation,  $s_{\theta_0}(X)$  is a binary statistic. Consequently, the same statistic is simultaneously optimal in the privacy and sampling regimes. By contrast, the logistic score takes a continuum of magnitudes, so its attaining statistic changes with  $\beta$  and its unified constant is not affine in  $\beta$ .

**5.2. Bounded mean estimation.** We next give a global minimax consequence of the local theory. Fix an integer  $p \geq 1$  and  $\ell < r$ , put  $\Delta = r - \ell$ , and let  $\mathcal{P}_{p, \ell, r} = \{P : \text{supp}(P) \subset [\ell, r]^p\}$ . The target is the mean  $\mu(P) = \mathbb{E}_P X \in \mathbb{R}^p$ , and the loss is squared Euclidean loss.

TABLE 2  
Efficiency constants for selected one-dimensional models.

| Model                            | Constant $\mathfrak{C}^{(\beta)}(q)$  | Attaining statistic                                                    |
|----------------------------------|---------------------------------------|------------------------------------------------------------------------|
| Bernoulli( $q$ )                 | $\beta q(1 - q) + (1 - \beta)/2$      | $x$                                                                    |
| Laplace location with scale $b$  | $b^2(2 - \beta)$                      | $\text{sign}(x - \theta_0)/2$                                          |
| Logistic location with scale $s$ | $6\beta s^2/(r_\beta(3 - r_\beta^2))$ | $\text{sign}(U) \min( U /r_\beta, 1)/2,$<br>$U = 2F_{\theta_0}(X) - 1$ |

PROPOSITION 5.2 (Bounded mean). Suppose  $n^2 \rho_n \rightarrow \infty$  and  $n \rho_n \rightarrow \lambda \in [0, \infty]$ . Then

$$(41) \quad \inf_{\hat{\mu}: \rho_n\text{-zCDP}} \sup_{P \in \mathcal{P}_{p, \ell, r}} \mathbb{E}_P \|\hat{\mu} - \mu(P)\|_2^2 \sim \Delta^2 \left( \frac{p}{4n} + \frac{p^2}{2\tau_n^2} \right).$$

The bound is attained by releasing  $\bar{X}_n + G_n$  with independent  $G_n \sim N_p(0, p\Delta^2 I_p/(2\tau_n^2))$ , where  $\bar{X}_n$  is the sample mean.

Thus, for distribution-free bounded mean estimation, the simple Gaussian release of the sample mean is globally asymptotically minimax, and no other private procedure can improve its first-order risk. The matching lower bound follows by restricting to the product Bernoulli submodel on the vertices of the hypercube; see Section S8.3 of the Supplementary Material (Cai and Li, 2026a).

5.3. *Gaussian mean.* We consider the multidimensional Gaussian mean model  $N_p(\theta, \sigma^2 I_p)$ , where  $p$  is fixed and  $\sigma^2$  is known. Let  $\chi_p = \|Z\|_2$  for  $Z \sim N_p(0, I_p)$ . For  $0 < \beta < 1$ , let  $b_{p, \beta} > 0$  be the unique solution of

$$(42) \quad \mathbb{E} \left( \frac{\chi_p}{b_{p, \beta}} - 1 \right)_+ = \frac{2p(1 - \beta)}{\beta}.$$

PROPOSITION 5.3 (Gaussian mean). For the Gaussian mean model, we have

$$(43) \quad \mathfrak{C}_\theta^{(\beta)}(I_p) = \frac{\beta p^2 \sigma^2}{\mathbb{E}[\chi_p \min(\chi_p, b_{p, \beta})]}, \quad \mathfrak{C}_\theta^{(0)}(I_p) = \frac{2\sigma^2 p^3}{(\mathbb{E}\chi_p)^2}, \quad \mathfrak{C}_\theta^{(1)}(I_p) = p\sigma^2.$$

For  $\beta = 0$  and  $0 < \beta < 1$ , the optimal statistic is given by the spatial sign and radial clipping respectively,

$$\Phi_{\theta_0, \text{priv}}(x) = \frac{1}{2} \frac{z}{\|z\|}, \quad \Phi_{\theta_0, \beta}(x) = \frac{1}{2b_{p, \beta}} \min \left( 1, \frac{b_{p, \beta}}{\|z\|_2} \right) z, \quad z = \frac{x - \theta_0}{\sigma}.$$

At  $\beta = 1$ , it is asymptotically given by a truncation with diverging radius.

The Gaussian mean model is the canonical example with an unbounded score. Under squared Euclidean loss, its location and rotational symmetries make the exact constant independent of the local center and reduce the multivariate optimization to the single clipping threshold  $b_{p, \beta}$ . The optimal statistic retains the score direction while clipping its radial magnitude: it is the spatial sign in the privacy regime, a radially clipped score in the transition regime, and has the linear score as its influence statistic in the sampling limit. Thus, unlike the binary score examples, the statistic optimal in the privacy regime is not sampling efficient, and the unified constant is not an affine interpolation of the constants in the privacy and sampling regimes.

**5.4. Gaussian quantile.** For Gaussian quantile estimation, if the variance is known, then it reduces to estimating the mean. However, when both the mean and variance are unknown, private estimation of the quantile is significantly different. Let  $z_\alpha$  denote the  $\alpha$ -quantile of the standard normal distribution, and consider the estimation of the quantile  $q_\alpha(\mu, \sigma) = \mu + z_\alpha \sigma$ . Fix a local center  $(\mu_0, \sigma_0)$  and take  $\Theta_n$  as a shrinking neighborhood. Define

$$(44) \quad d_\alpha = \inf_{\lambda \in \mathbb{R}} \mathbb{E} |Z - \lambda(Z^2 - 1 - z_\alpha Z)|, \quad Z \sim N(0, 1).$$

**PROPOSITION 5.4 (Gaussian quantile with unknown scale).** Suppose that  $\alpha \in (0, 1)$ ,  $\sigma_0 > 0$ ,  $n^2 \rho_n \rightarrow \infty$ , and  $n \rho_n \rightarrow 0$ . For the Gaussian quantile problem, the privacy constant is

$$\lim_{n \rightarrow \infty} \tau_n^2 \inf_{\hat{q}_\alpha: \rho_n\text{-ZCDP}} \sup_{(\mu, \sigma) \in \Theta_n} \mathbb{E}_{\mu, \sigma} (\hat{q}_\alpha - q_\alpha(\mu, \sigma))^2 = \frac{2\sigma_0^2}{d_\alpha^2},$$

where  $d_\alpha$  is defined in (44). If  $\lambda_\alpha^*$  minimizes (44), the optimal statistic is generated by

$$\phi_\alpha^*(x) = \frac{1}{2} \text{sign} \left( Z_0 - \lambda_\alpha^* (Z_0^2 - 1 - z_\alpha Z_0) \right), \quad Z_0 = \frac{x - \mu_0}{\sigma_0}.$$

Here  $H_\alpha(Z) = Z^2 - 1 - z_\alpha Z$  is the scaled score in the local nuisance direction that leaves  $q_\alpha$  unchanged. The quantity  $d_\alpha$  is the  $L^1$  residual after the location score  $Z$  is adjusted for this nuisance direction, and the resulting sign statistic has zero covariance with  $H_\alpha(Z)$ . At  $\alpha = 1/2$ , symmetry gives  $\lambda_\alpha^* = 0$ , so the statistic reduces to  $\text{sign}(Z)/2$  and the constant to  $\pi\sigma_0^2$ , recovering the unknown-scale Gaussian mean problem. In comparison, the non-private optimal estimator is given by the plug-in MLE  $\bar{X} + z_\alpha \hat{\sigma}$ . This example further shows differences between optimal private and non-private estimation.

**5.5. Finite alphabet and categorical model.** Consider the finite alphabet model

$$\mathcal{X} = \{x_1, \dots, x_m\}, \quad P_\theta(X = x_i) = \pi_i(\theta) > 0,$$

on a neighborhood of  $\theta_0$ . Under this model, the optimization for efficiency constant becomes finite dimensional and can be computed by a semidefinite programming. Due to its complexity, we only present the simpler form in the privacy regime, and defer the general form to the supplementary material. At  $\theta_0$ , write  $\pi = (\pi_1, \dots, \pi_m)^\top$ ,  $S \in \mathbb{R}^{m \times p}$  with  $S_i = s_{\theta_0}(x_i)^\top$ , and  $B = \text{diag}(\pi)S$ .

**PROPOSITION 5.5 (Finite alphabet SDP).** If  $I(\theta_0) \succ 0$  and  $W \succ 0$ , then

$$(45) \quad \begin{aligned} \mathfrak{C}_{\theta_0}(W) &= \frac{1}{2} \min_{G \in \mathbb{S}_+^p, Z \in \mathbb{S}_+^p} \text{Tr } Z \\ \text{subject to } G\pi &= 0, \quad G_{ii} + G_{jj} - 2G_{ij} \leq 1, \quad \begin{pmatrix} B^\top G B & W^{1/2} \\ W^{1/2} & Z \end{pmatrix} \succeq 0. \end{aligned}$$

The minimum of (45) is attained.

The optimization problem in Proposition 5.5 not only gives the exact efficiency constant, but also implies the construction of the optimal statistics. In detail, for any optimal  $G$  solving (45) with factorization  $G = FF^\top$ , choose an orthonormal basis  $Q$  of  $\text{ran}(F^\top B)$ , the optimal statistic is given by  $\Phi(x_i) = Q^\top F_i^\top$ . Conversely, any optimal statistic corresponds to an optimal  $G$ .

The SDP gives a finite-dimensional form of the loss-specific incompatibility in Proposition 2.15. Its feasible Gram region is fixed by the model, whereas the loss matrix  $W$  enters

the objective and may select a different optimal  $G$ . If two losses have no common minimizing Gram matrix, no single statistic can be efficient for both.

As a special case, the categorical probability vector model further simplifies the computation of the exact constant. Let  $\Delta_{k-1} = \{q = (q_1, \dots, q_k) \in \mathbb{R}^k : q_j \geq 0, \sum_{j=1}^k q_j = 1\}$  be the probability simplex of dimension  $k-1$ . Let  $q$  denote the probability vector to be estimated, assumed to be an interior point of the simplex. Use category  $k$  as the baseline and put  $d = k-1$ . The sampling covariance matrix of the first  $d$  category indicators is

$$V_q = \text{diag}(q_1, \dots, q_d) - q_{1:d} q_{1:d}^\top.$$

Define the Gram region generated by diameter-one statistics

$$\mathcal{G}_k = \{J \succeq 0 : J_{aa} \leq 1, \quad J_{aa} + J_{bb} - 2J_{ab} \leq 1 \text{ for all } a, b \in \{1, \dots, d\}\}.$$

PROPOSITION 5.6 (Categorical model). For  $W \succ 0$ ,

$$(46) \quad \begin{aligned} \mathfrak{C}_q(W) &= \frac{1}{2} \min_{J \in \mathcal{G}_k, J \succ 0} \text{Tr}(WJ^{-1}), \\ \mathfrak{C}_q^{(\beta)}(W) &= \beta \text{Tr}(WV_q) + (1 - \beta) \mathfrak{C}_q(W), \quad 0 \leq \beta \leq 1. \end{aligned}$$

Every statistic that is optimal in the privacy regime remains optimal for  $0 \leq \beta < 1$ . Under squared Euclidean loss on the full probability vector,  $W = I_d + \mathbf{1}_d \mathbf{1}_d^\top$ , and

$$\mathfrak{C}_q^{(\beta)} = \beta(1 - \|q\|_2^2) + (1 - \beta)(k - 1).$$

Here,  $\text{Tr}(WV_q)$  is the sampling regime constant dictated by the Fisher information, whereas  $\mathfrak{C}_q(W)$  is the privacy regime constant and depends only on  $W$  and  $k$ . The categorical model is exceptional because, for each fixed loss, a statistic optimal in the privacy regime remains optimal throughout  $0 \leq \beta < 1$ . Thus, unlike in a general finite-alphabet model, the same bounded statistic jointly retains the sampling and privacy information needed in every non-sampling regime.

For squared Euclidean loss on the full probability vector, the scaled one-hot statistic  $\Phi_{\text{hist}}(j) = e_j / \sqrt{2}$ , up to translation to the simplex tangent space, has diameter one and attains the constant. The term  $1 - \|q\|_2^2$  is the sampling contribution of the empirical histogram, whereas  $k-1$  is the privacy contribution from the  $(k-1)$ -dimensional simplex tangent space. The sampling contribution is largest at the uniform probability vector.

5.6. *Generalized linear regression.* We consider the canonical generalized linear model (GLM) with Gaussian design

$$(47) \quad X \sim N_p(0, I_p), \quad p_\vartheta(y | x) = \exp \left( \frac{yx^\top \vartheta - b(x^\top \vartheta)}{\kappa} + c(y, \kappa) \right),$$

where  $\vartheta \in \mathbb{R}^p$  is the regression parameter,  $X \in \mathbb{R}^p$  is the covariate, and  $Y$  is the response. The function  $b$  is the cumulant function,  $\kappa > 0$  is a known dispersion parameter, and  $c(y, \kappa)$  is the normalizing term.

The results for GLM under Gaussian design resemble that of Gaussian mean estimation. Let  $Y_0$  follow the response family at natural parameter zero and, independently, let  $\chi_p = \|Z\|_2$  for  $Z \sim N_p(0, I_p)$ .

$$(48) \quad A_0 = \frac{Y_0 - b'(0)}{\kappa}, \quad d_0 = \mathbb{E}_0 |A_0|, \quad R_0 = |A_0| \chi_p,$$

At the null center, the score is  $s_0(X, Y_0) = A_0 X$ . Suppose [Assumptions 1](#) and [2](#) hold and  $b''(0) > 0$ . For  $0 < \beta < 1$ , let  $t_\beta > 0$  be the unique solution of  $\mathbb{E}(R_0/t_\beta - 1)_+ = 2p(1 - \beta)/\beta$ .

PROPOSITION 5.7 (GLM with Gaussian design). For the GLM (47) at  $\vartheta_0 = 0$ , the exact local constant under squared Euclidean loss is

$$(49) \quad \mathfrak{C}_0^{(\beta)}(I_p) = \frac{\beta p^2}{\mathbb{E}[R_0 \min(R_0, t_\beta)]}, \quad \mathfrak{C}_0^{(0)}(I_p) = \frac{2p^3}{(\mathbb{E}\chi_p)^2 d_0^2}, \quad \mathfrak{C}_0^{(1)}(I_p) = \frac{p\kappa}{b''(0)}.$$

For  $\beta = 0$  and  $0 < \beta < 1$ , the optimal statistic is given by the score direction and radial clipping, respectively,

$$\Phi_{\text{priv}}(x, y) = \frac{1}{2} \frac{s_0(x, y)}{\|s_0(x, y)\|_2}, \quad \Phi_\beta(x, y) = \frac{1}{2} \frac{s_0(x, y)}{\|s_0(x, y)\|_2} \min\left(\frac{\|s_0(x, y)\|_2}{t_\beta}, 1\right),$$

with both statistics set to zero when  $s_0(x, y) = 0$ . At  $\beta = 1$ , an asymptotically efficient statistic is given by a score truncation with diverging radius.

TABLE 3  
Privacy and sampling regime constants for selected GLMs with Gaussian design at the null center.

| Model                                                     | $R_0$                                             | Privacy                                    | Sampling     |
|-----------------------------------------------------------|---------------------------------------------------|--------------------------------------------|--------------|
| Gaussian linear regression with noise variance $\sigma^2$ | $\chi_1 \chi_p / \sigma$                          | $\pi \sigma^2 p^3 (\mathbb{E}\chi_p)^{-2}$ | $p \sigma^2$ |
| Logistic regression                                       | $\chi_p / 2$                                      | $8 p^3 (\mathbb{E}\chi_p)^{-2}$            | $4p$         |
| Poisson log-linear regression                             | $ Y_0 - 1  \chi_p, \\ Y_0 \sim \text{Poisson}(1)$ | $2^{-1} e^2 p^3 (\mathbb{E}\chi_p)^{-2}$   | $p$          |

At the null center, Gaussian design makes the score spherically symmetric with radial magnitude  $R_0$ . As in Gaussian mean estimation, the multivariate calculation therefore reduces to the scalar threshold  $t_\beta$ . The same reduction applies to GLMs with spherically symmetric designs. Table 3 lists the constants for several GLM examples.

## 6. Extensions.

6.1. *Gaussian differential privacy.* Our efficiency theory naturally extends to the Gaussian DP notion (Dong, Roth and Su, 2022). For probability measures  $P$  and  $Q$ , define the trade-off function

$$T(P, Q)(a) = \inf_{\varphi: \mathbb{E}_P \varphi \leq a} (1 - \mathbb{E}_Q \varphi), \quad a \in [0, 1],$$

where the infimum is over measurable tests  $\varphi$  taking values in  $[0, 1]$ .

DEFINITION 6.1 ( $\mu$ -GDP). A mechanism  $M$  with a standard Borel output space is  $\mu$ -Gaussian differentially private (GDP) if

$$T(M(S), M(S')) \geq G_\mu := T(N(0, 1), N(\mu, 1))$$

pointwise on  $[0, 1]$  for every adjacent pair  $S, S'$ .

We use the following standard comparison, Gaussian mechanism, and composition properties of GDP (Dong, Roth and Su, 2022, Theorems 1, 2, and 4 and Corollary 2).

LEMMA 6.2. *The following statements hold.*

- (i) Every  $\mu$ -GDP mechanism is  $(\mu^2/2)$ -zCDP.

(ii) The Gaussian mechanism in [Lemma 3.1](#) is  $\mu$ -GDP when  $\rho = \mu^2/2$ , and the adaptive composition of  $\mu_j$ -GDP mechanisms,  $j = 1, \dots, k$ , is  $(\sum_{j=1}^k \mu_j^2)^{1/2}$ -GDP.

The zCDP theory transfers directly to GDP thanks to the connection between the two and the construction of the attaining estimator. On one hand, [Lemma 6.2\(i\)](#) makes the lower bound immediate by inclusion. On the other hand, by [Lemma 6.2\(ii\)](#), the moment-map inverse estimator and preliminary localization used in the upper bounds also satisfy GDP. Combining the upper and lower part, we obtain the following efficiency corollary for GDP.

**COROLLARY 6.3 (Efficiency under GDP).** *Let  $(\mu_n)$  be a positive sequence and set  $\rho_n = \mu_n^2/2$ . In the notation of [\(13\)](#) and [\(18\)](#), the following conclusions hold.*

(i) Under the assumptions of [Theorem 2.8](#),

$$(50) \quad \lim_{n \rightarrow \infty} \sigma_n^{-2} \inf_{\hat{\theta}: \mu_n\text{-GDP}} \sup_{\theta \in \tilde{\Theta}_n} \mathbb{E}_\theta \|\hat{\theta} - \theta\|_W^2 = \mathfrak{C}_{\theta_0}^{(\beta)}(W).$$

(ii) Under the assumptions of [Theorem 2.12](#),

$$(51) \quad \lim_{n \rightarrow \infty} \inf_{\hat{\theta}: \mu_n\text{-GDP}} \sup_{\theta \in K} \frac{\sigma_n^{-2} \mathbb{E}_\theta \|\hat{\theta} - \theta\|_W^2}{\mathfrak{C}_\theta^{(\beta)}(W)} = 1.$$

**6.2. Differentiable targets.** Although our results are stated for  $\theta$  itself, the same theory applies to a differentiable target  $\psi : \Theta \rightarrow \mathbb{R}^m$  under quadratic loss  $\|\hat{\psi} - \psi(\theta)\|_W^2$  for a quadratic loss matrix  $W \succ 0$ . The optimal constant  $\mathfrak{C}_{\psi, \theta}^{(\beta)}(W)$  is obtained via the local expansion of  $\psi$  and is given by

$$(52) \quad \mathfrak{C}_{\psi, \theta}^{(\beta)}(W) = \mathfrak{C}_\theta^{(\beta)}(W_{\psi, \theta}), \quad W_{\psi, \theta} = D\psi(\theta)^\top W D\psi(\theta).$$

Detailed formal results are given in Section S6.10 of the Supplementary Material ([Cai and Li, 2026a](#)).

**7. Conclusion.** This paper develops an exact asymptotic efficiency theory for estimation under central zCDP. Privacy does not simply apply a scalar discount to ordinary Fisher information, but introduces a set-valued information region recording the privacy constraint. The same region supplies both sides of the theory: it constrains the Fisher information of every zCDP output experiment and generates matching moment-map inverse estimators based on nearly optimal statistics. The set-valued information formulation extends across privacy regimes and shows how privacy and sampling interact through the same bounded statistic. In particular, in the transition regime, the statistic's sampling covariance and privacy sensitivity yield a joint variational constant rather than a weighted average of the two endpoint constants, while the sampling regime limit recovers ordinary Fisher efficiency.

Our theory also supplies practically useful private estimators. For each loss and privacy regime, a moment-map inverse estimator based on a nearly optimal statistic and its Gaussian release attains the optimal constant, so no nonadditive, nonlinear, or otherwise unrestricted zCDP mechanism can improve the leading constant for these problems. In the examples, this principle yields explicit constructions, showing how the information geometry translates into concrete private procedures.

Several directions now become natural. An immediate question is to develop optimal private estimators for other parametric models and numerical methods for optimizing the information region and its associated mechanisms. A natural direction is to develop an exact efficiency theory under  $(\varepsilon, \delta)$ -differential privacy, where the privacy constraint calls for a

different geometric characterization of the attainable local experiments. More broadly, semi-parametric functionals and covariance or eigenspace problems offer settings in which nuisance structure and matrix geometry may produce genuinely new private efficiency phenomena. Establishing efficiency theory for nonparametric estimation under privacy is also an interesting future problem.

We believe that the theory developed here provides a foundation for a general theory of statistical efficiency under differential privacy, in which privacy, statistical information, and decision loss are understood through a common information geometry.

## SUPPLEMENTARY MATERIAL

### Supplement to “Exact Asymptotic Efficiency under zCDP: Diameter-Constrained Information Geometry”

The supplement provides complete proofs of the main results, including information contraction, sharp lower bounds, and attainment by moment-map inverse estimators based on Gaussian releases. It also develops reusable model reductions, and detailed calculations of exact constants and attaining procedures across sampling and privacy regimes.

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